

Digital Systems

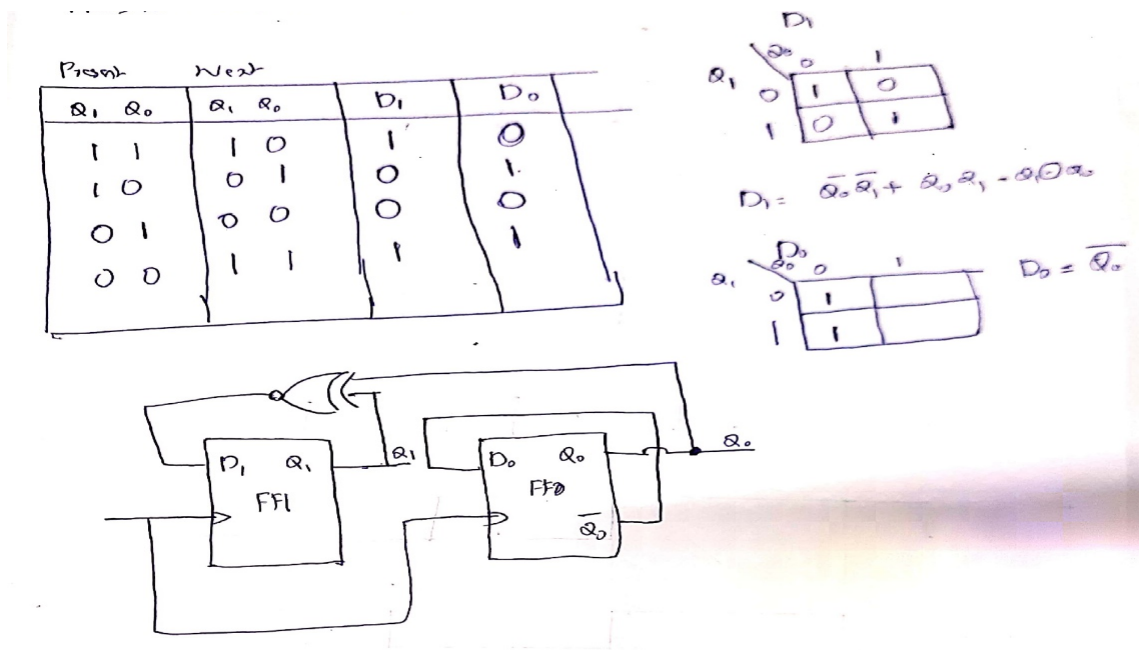
Tutorial 7

March-June 2023

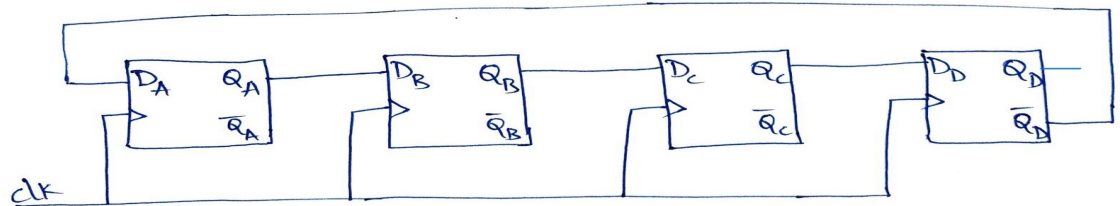
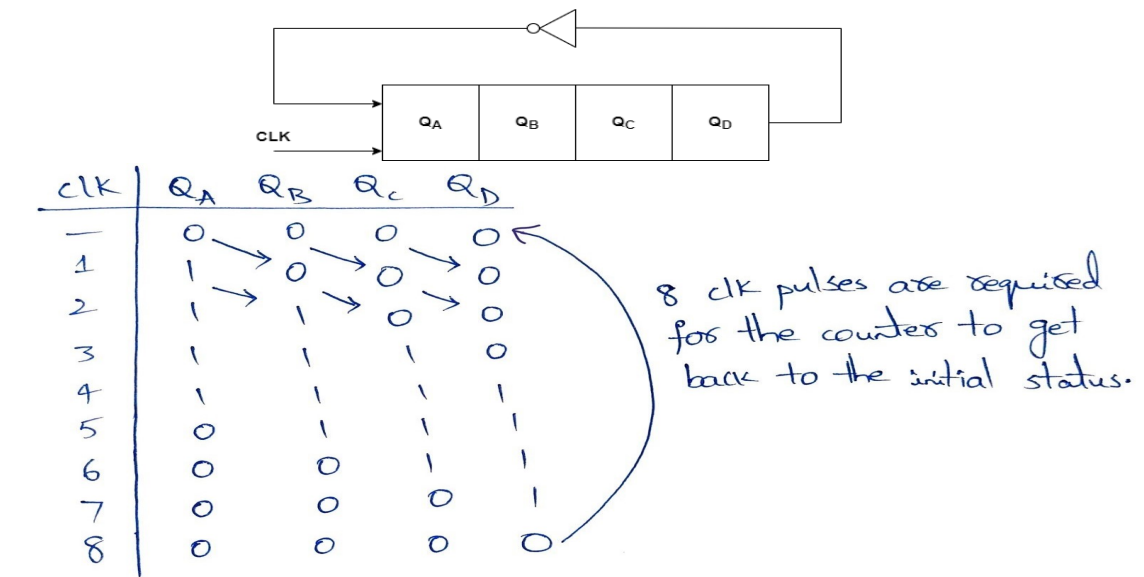
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- Design a synchronous counter which can go through the following sequence of states using D-flipflops.

11 → 10 → 01 → 00 → 11 → 10 → 01 → 00.....



- Find the number of clock pulses does the given counter require to get back to the initial status. Give the expression for output frequency of the counter. Implement the counter using D-flipflops. Also give the generalized name of the counter.



→ The counter is called "Johnson counter". It is also known as "twisted ring counter".

Let (input) clock frequency be f_i .
Therefore, output frequency $f_0 = f_i / \text{no. of clk} = f_i / 8$.

- Design a mod-10 synchronous sequential counter using J-K flipflops.

Present state				Next state				Excitation table							
Q_4	Q_3	Q_2	Q_1	Q_4^*	Q_3^*	Q_2^*	Q_1^*	J_4	K_4	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	0	0	0	1	0	x	0	x	0	x	1	x
0	0	0	1	0	0	1	0	0	x	0	x	1	x	x	1
0	0	1	0	0	0	1	1	0	x	0	x	x	0	1	x
0	0	1	1	0	1	0	0	0	x	1	x	x	1	x	1
0	1	0	0	0	1	0	1	0	x	x	0	0	x	1	x
0	1	0	1	0	1	1	0	0	x	x	0	1	x	x	1
0	1	1	0	0	1	1	1	0	x	x	0	x	0	1	x
0	1	1	1	1	0	0	0	1	x	x	1	x	1	x	1
1	0	0	0	1	0	0	1	x	0	0	x	0	x	1	x
1	0	0	1	0	0	0	0	x	1	0	x	0	x	x	1

J_4

$Q_3 Q_1$	00	01	11	10
$Q_4 Q_2$	00			
01			1	
11	x	x	x	x
10	x	x	x	x

$$J_4 = Q_3 Q_2 Q_1$$

K_4

$Q_3 Q_1$	00	01	11	10
$Q_4 Q_2$	00	x	x	x
01	x	x	x	x
11	x	x	x	x
10		1	x	x

$$K_4 = Q_1$$

J_3

$Q_2 Q_1$	00	01	11	10
$Q_4 Q_3$				
00				
01	x	x	x	x
11	x	x	x	x
10				

$$J_3 = Q_2 Q_1$$

K_3

$Q_2 Q_1$	00	01	11	10
$Q_4 Q_3$				
00	x	x	x	x
01	x	x	x	x
11	x	x	x	x
10	x	x	x	x

$$K_3 = Q_2 Q_1$$

J_2

$Q_3 Q_1$	00	01	11	10
$Q_4 Q_2$				
00				
01	1	x	x	
11	1	x	x	
10	x	x	x	x

$$J_2 = \bar{Q}_4 Q_1$$

K_2

$Q_3 Q_1$	00	01	11	10
$Q_4 Q_2$				
00	x	x	1	
01	x	x	1	
11	x	x	x	x
10	x	x	x	x

$$K_2 = Q_1$$

J_1

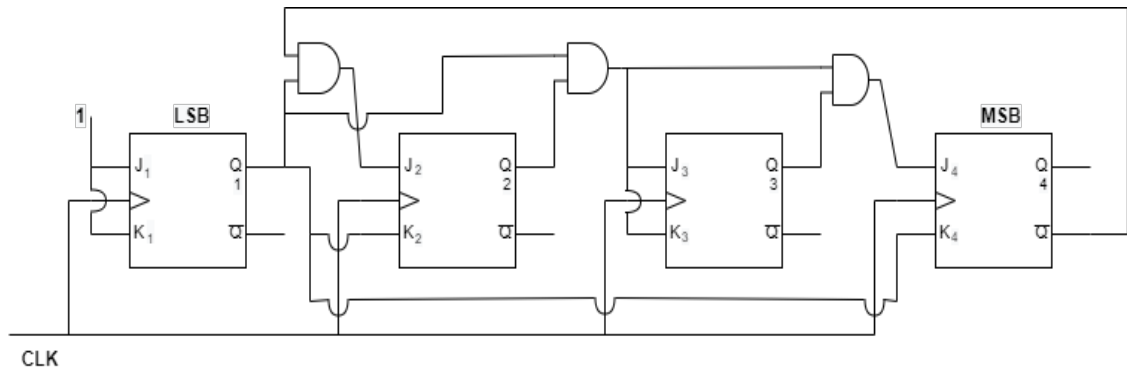
$Q_2 Q_1$	00	01	11	10
$Q_4 Q_3$				
00	1	x	x	1
01	1	x	x	1
11	x	x	x	x
10	1	x	x	x

$$J_1 = 1$$

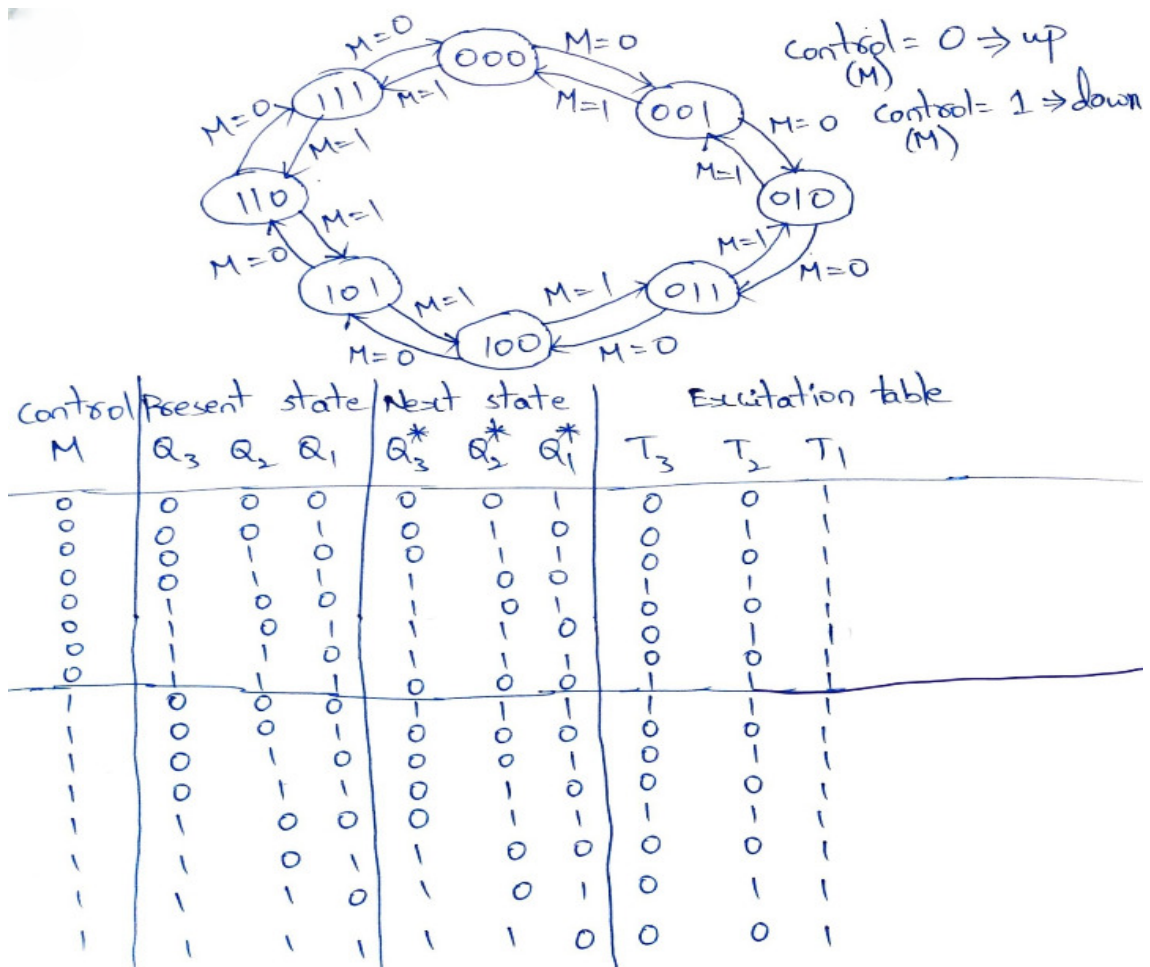
K_1

$Q_2 Q_1$	00	01	11	10
$Q_4 Q_3$				
00	x	1	1	x
01	x	1	1	x
11	x	x	x	x
10	x	1	x	x

$$K_1 = 1$$



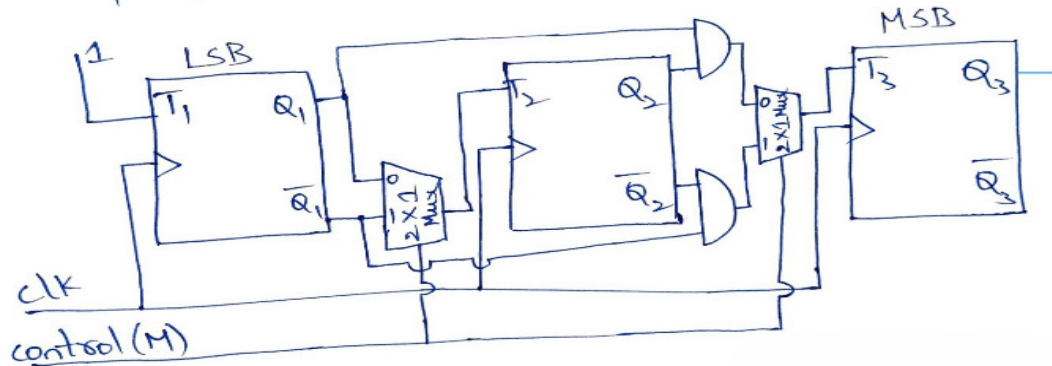
4. Design a 3-bit synchronous up-down counter using T-flipflops. Use multiplexers for up/down counting selection.



$$T_3 = \overline{M}\overline{Q}_3Q_2Q_1 + \overline{M}Q_3Q_2Q_1 + \underbrace{M\overline{Q}_3\overline{Q}_2\overline{Q}_1 + MQ_3\overline{Q}_2\overline{Q}_1}_{= \overline{M}Q_2Q_1 + M\overline{Q}_2\overline{Q}_1}$$

$$T_2 = \overline{M}\overline{Q}_3\overline{Q}_2Q_1 + \overline{M}\overline{Q}_3Q_2Q_1 + \overline{M}Q_3\overline{Q}_2Q_1 + \overline{M}Q_3Q_2Q_1 + \underbrace{M\overline{Q}_3\overline{Q}_2\overline{Q}_1 + M\overline{Q}_3\overline{Q}_2Q_1 + MQ_3\overline{Q}_2\overline{Q}_1}_{= \overline{M}Q_1 + M\overline{Q}_1 = M \oplus Q_1}$$

$$T_1 = 1$$



5. Find the number of flipflops required to design a mod-16 counter in terms of (i) ripple counter, (ii) twisted ring counter and (iii) ring counters.

ii. Mod-16 counter counts 16 clk pulses.

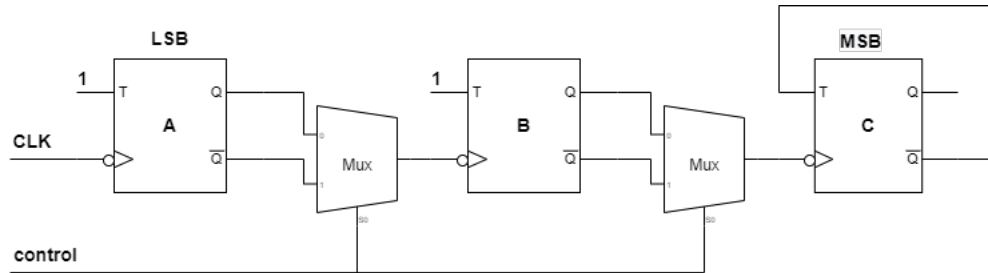
i) ripple counter: counts 2^n clk pulses.

$$2^n = 16 \Rightarrow n = \underline{4} \text{ f/f s.}$$

ii) twisted ring counter: counts $2n$ clk pulses
(Johnson) $2n = 16 \Rightarrow n = \underline{8} \text{ f/f s.}$

iii) ring counter: counts n clk pulses
 $n = \underline{16} \text{ f/f s.}$

6. State the required condition for the below given counter to work as down counter. Assuming initial condition 000, find the number of clock pulses required to get initial condition again in the view of down counter.



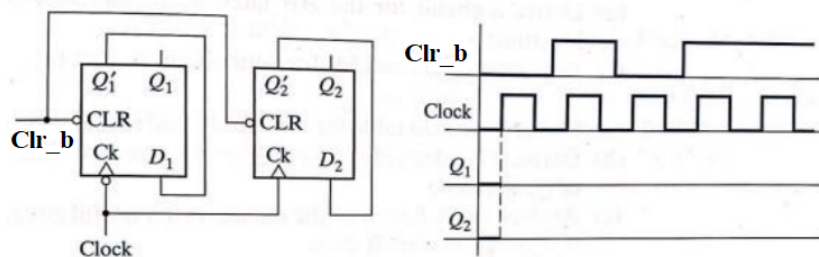
When $\text{control} = 1$, this counter acts as down counter.
(so that $\bar{Q}_A$ & $\bar{Q}_B$ will be connected as negative edge triggered clock).

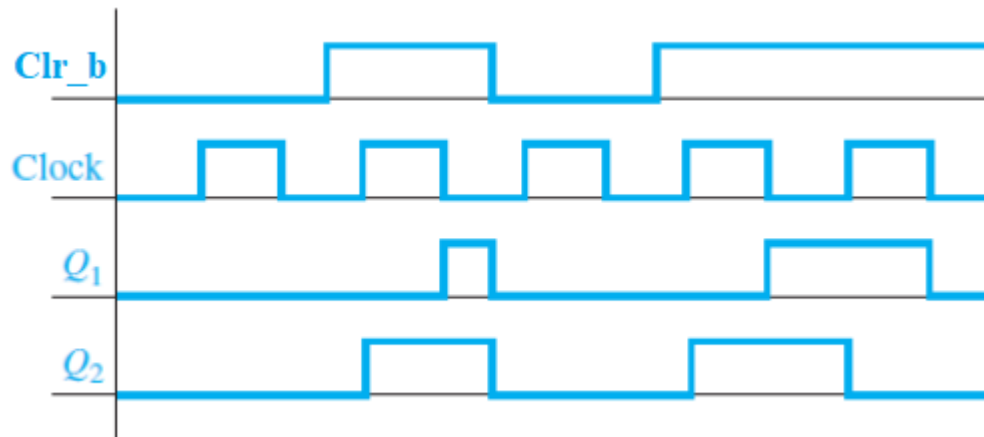
clk	Q_C	Q_B	Q_A
—	0	0	0
1	1	1	1
2	1	1	0
3	1	0	1
4	1	0	0
5	1	1	1

→ Till 4 clk pulses, it is down counting.

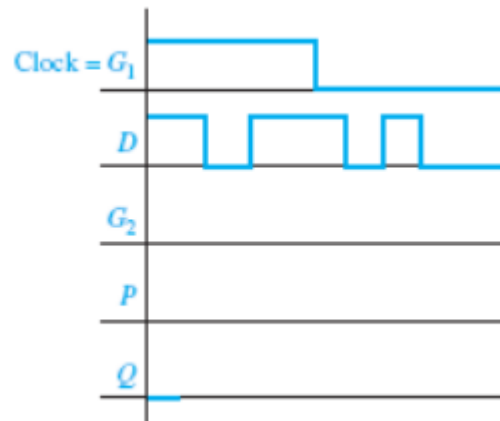
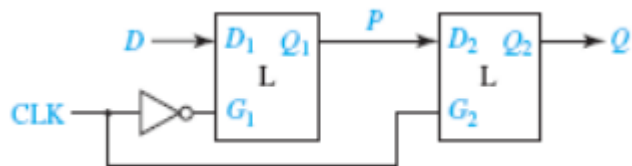
→ From 5th clk, particular down counting sequence repeats. But initial condition can never be achieved.

7. Complete the timing diagram for the following circuit. Note that the Clk inputs on the two flip-flops are different.

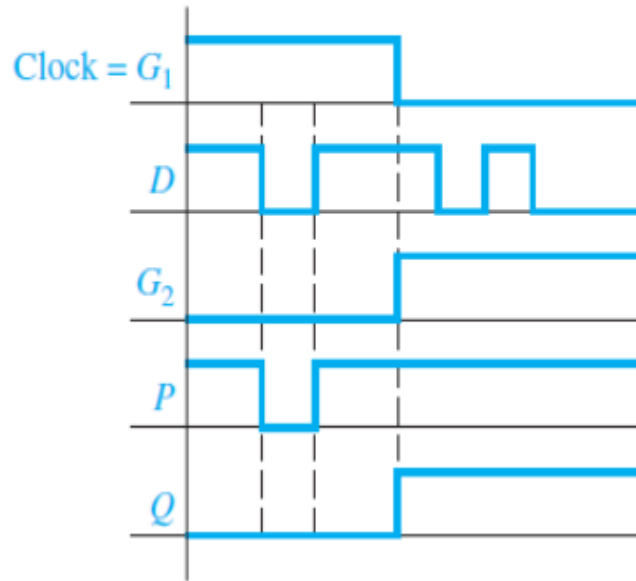




8. What change must be made to the circuit given below to implement a falling-edge-triggered D flip-flop? Complete the given timing diagram for the modified flip-flop.



Connect the clock directly to the input G_1 and connect the clock to G_2 through an inverter.



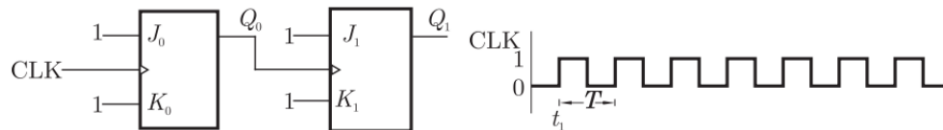
9. A flip-flop has a 3 ns delay from the time the clock edge occurs to the time the output is complemented. What is the maximum delay in a 10-bit binary ripple counter that uses these flip-flops? What is the maximum frequency at which the counter can operate reliably?

Solution: The worst case is when all 10 flip-flops are complemented.

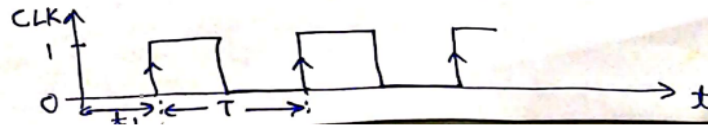
The maximum delay is $10 \times 3 \text{ ns} = 30 \text{ ns}$.

The maximum frequency = $1/\text{max_delay} = 10^9/30 = 33.3 \text{ MHz}$.

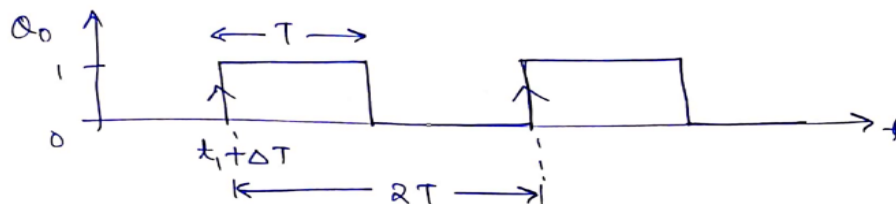
10. For each of the positive-edge triggered JK-FF used in the figure below, the propagation delay is ΔT . Draw the output waveform at Q_1 .



Since the inputs to both JK flip-flops is 11, so both FFs are in Toggle mode and act as T-FF i.e. the output will change ~~with~~ (flip) with each clock pulse. Given that the input to the CLK is:



The output of first FF: Q_0 occurs after time ΔT as shown below:



This so happens because the FFs are positive edge-triggered, and thus, only respond to the positive edge of the clock.

The output Q of second FF occurs after time ΔT when gets input (i.e. after ΔT from $t_1 + \Delta T$) as shown below



11. A digital system has a clock generator that produces pulses at a frequency of 80 MHz. But I have another digital circuit which requires a clock of frequency 20 MHz (or 50 ns). Is it possible to run the second circuit with some modification on the first circuit?...If possible suggest the method.

Solution:

The clock generator has a period of 12.5 ns. Use a 2-bit counter to count four pulses.

$$80/4 = 20 \text{ MHz; cycle time} = 1000 \times 10^{-9} / 20 = 50 \text{ ns.}$$

Practice Questions:

- Design a 4-bit binary counter using S-R flipflops for counting even numbers only.

Present state				Next state				Excitation table							
Q_4	Q_3	Q_2	Q_1	Q_4^*	Q_3^*	Q_2^*	Q_1^*	S_4	R_4	S_3	R_3	S_2	R_2	S_1	R_1
0	0	0	0	0	0	1	0	0	x	0	x	1	0	0	x
0	0	1	0	0	1	0	0	0	x	1	0	0	1	0	x
0	1	0	0	0	1	1	0	0	x	x	0	1	0	0	x
0	1	1	0	1	0	0	0	1	0	0	1	0	1	0	x
1	0	0	0	1	0	1	0	x	0	0	x	1	0	0	x
1	0	1	0	1	1	0	0	x	0	1	0	0	1	0	x
1	1	0	0	1	1	1	0	x	0	x	0	1	0	0	x
1	1	1	0	0	0	0	0	0	1	0	1	0	1	0	x

S_4

$Q_2 \backslash Q_1$	00	01	11	10
$Q_4 \backslash Q_3$		x	x	
00		x	x	1
01		x	x	
11	x	x	x	
10	x	x	x	x

$S_4 = \bar{Q}_4 Q_3 Q_2$

R_4

$Q_2 \backslash Q_1$	00	01	11	10
$Q_4 \backslash Q_3$	x	x	x	x
00	x	x	x	
01	x	x	x	1
11		x	x	
10		x	x	

$R_4 = Q_4 Q_3 Q_2$

S_3

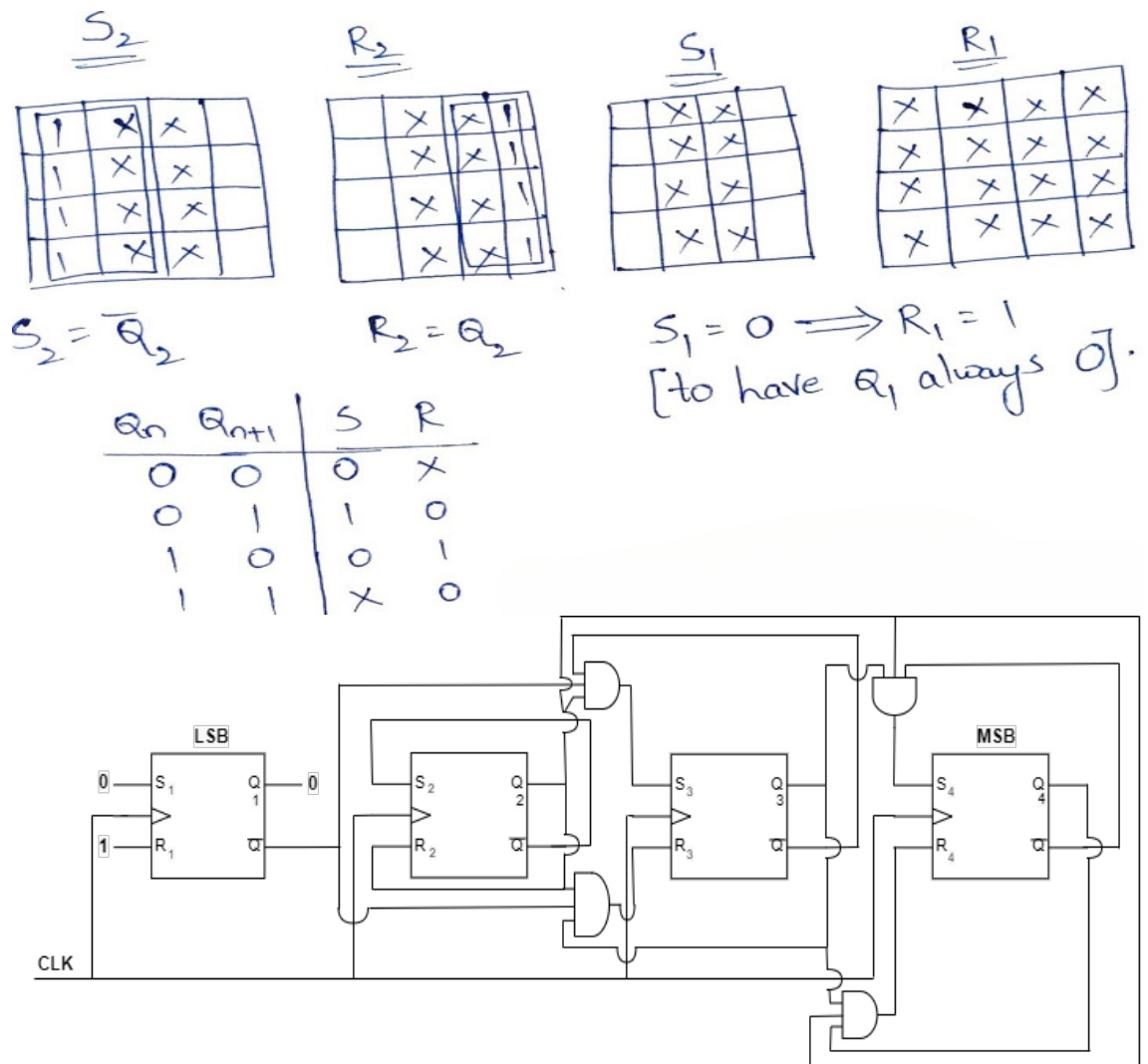
$Q_2 \backslash Q_1$	00	01	11	10
$Q_4 \backslash Q_3$		x	x	1
00		x	x	
01		x	x	
11	x	x	x	
10		x	x	1

$S_3 = \bar{Q}_3 Q_2 \bar{Q}_1$

R_3

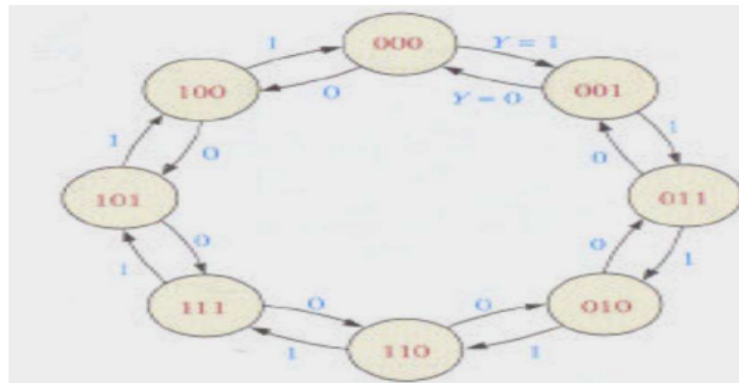
$Q_2 \backslash Q_1$	00	01	11	10
$Q_4 \backslash Q_3$	x	x	x	
00	x	x	x	1
01		x	x	
11		x	x	
10	x	x	x	

$R_3 = Q_3 Q_2 \bar{Q}_1$



2. Design a synchronous 3-bit Up/Down counter having Gray code sequence using JK flipflops.

Step 1= State Diagram



Step 2 = Next State Table

Present State			Next State					
			Y = 0 (DOWN)			Y = 1 (UP)		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	0	1	0	0	0	0	1
0	0	1	0	0	0	0	1	1
0	1	1	0	0	1	0	1	0
0	1	0	0	1	1	1	1	0
1	1	0	0	1	0	1	1	1
1	1	1	1	1	0	1	0	1
1	0	1	1	1	1	1	0	0
1	0	0	1	0	1	0	0	0

$Y = \overline{\text{UP/DOWN}}$ control input.

Step 3= Slecting JK flip-flops and finding values of J & K by Karnaugh Maps:

Present State Q_2, Q_1, Q_0	Next State Q_2, Q_1, Q_0	J_2	K_2	J_1	K_1	J_0	K_0
0 0 0	1 0 0	1	x	0	x	0	x
0 0 1	0 0 0	0	x	0	x	x	1
0 1 1	0 0 1	0	x	x	1	x	0
0 1 0	0 1 1	0	x	x	0	1	x
1 1 0	0 1 0	x	1	x	0	0	x
1 1 1	1 1 0	x	0	x	0	x	1
1 0 1	1 1 1	x	0	1	x	x	0
1 0 0	1 0 1	x	0	0	x	1	x

$Y = 0$ (DOWN)

J_2

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	1	0	0	0
1	x	x	x	x

$$J_2 = \bar{Q}_1 \bar{Q}_0 Y$$

K_2

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	x	x	x	x
1	0	0	0	1

$$K_2 = Q_1 \bar{Q}_0 Y$$

J_1

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	0	0	x	x
1	0	1	x	x

$$J_1 = Q_2 Q_0 Y$$

K_1

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	x	x	1	0
1	x	x	0	0

$$K_1 = \bar{Q}_2 Q_0 Y$$

J_0

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	0	x	x	1
1	1	x	x	0

$$J_0 = (Q_2 \bar{Q}_1 + \bar{Q}_2 Q_1) Y$$

K_0

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	x	1	0	x
1	x	0	1	x

$$K_0 = (\bar{Q}_2 \bar{Q}_1 + Q_2 Q_1) Y$$

Present State Q_2, Q_1, Q_0	Next State Q_2, Q_1, Q_0	J_2	K_2	J_1	K_1	J_0	K_0
0 0 0	0 0 1	0	x	0	x	1	x
0 0 1	0 1 1	0	x	1	x	x	0
0 1 1	1 1 1	0	x	x	0	0	x
0 1 0	1 1 0	0	x	x	0	1	x
1 1 0	1 0 1	x	0	x	1	x	0
1 1 1	1 0 0	x	0	0	x	x	1
1 0 1	1 0 0	x	0	0	x	0	x
1 0 0	0 0 0	x	1	0	x	0	x

$Y = 1$ (UP)

J_2

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	0	0	0	1
1	x	x	x	x

$$J_2 = Q_1 \bar{Q}_0 Y$$

K_2

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	x	x	x	x
1	1	0	0	0

$$K_2 = \bar{Q}_1 \bar{Q}_0 Y$$

J_1

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	0	1	x	x
1	0	0	x	x

$$J_1 = \bar{Q}_2 Q_0 Y$$

K_1

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	x	x	0	0
1	x	x	1	0

$$K_1 = Q_2 Q_0 Y$$

J_0

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	1	x	x	0
1	0	x	x	1

$$J_0 = (\bar{Q}_2 \bar{Q}_1 + Q_2 Q_1) Y$$

K_0

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	1	0	1	x
1	x	1	0	x

$$K_0 = (Q_2 \bar{Q}_1 + \bar{Q}_2 Q_1) Y$$

Combining the equations

$$J_2 = \bar{Q}_1 \bar{Q}_0 \bar{Y} + Q_1 \bar{Q}_0 Y = \bar{Q}_0 [Q_1 \bar{Y} + \bar{Q}_1 Y] = \bar{Q}_0 [Q_1 \oplus Y]$$

$$J_1 = Q_2 \bar{Q}_0 \bar{Y} + \bar{Q}_2 Q_0 Y = Q_0 [Q_2 \bar{Y} + \bar{Q}_2 Y] = \underline{Q_0 (Q_2 \oplus Y)}$$

$$J_0 = (Q_2 \bar{Q}_1 + \bar{Q}_2 Q_1) \bar{Y} + (Q_2 \bar{Q}_1 + \bar{Q}_2 Q_1) Y$$

$$= (Q_2 \oplus Q_1) \bar{Y} + (Q_2 \oplus Q_1) Y = \underline{Q_1 \oplus Q_2 \oplus Y}$$

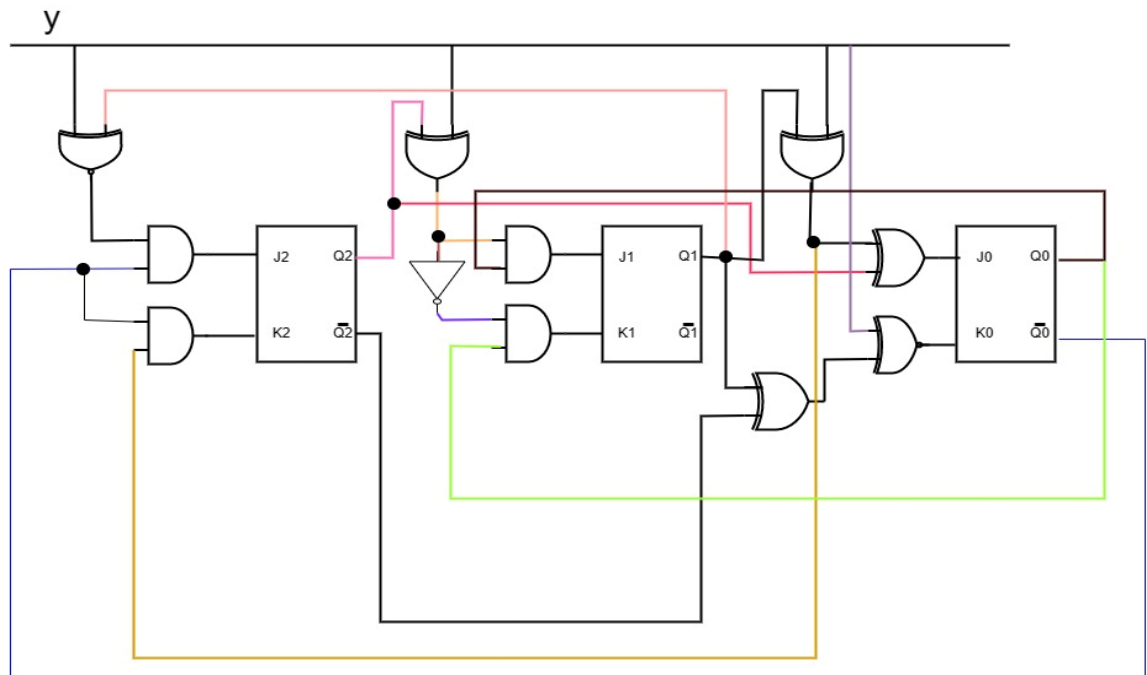
$$K_2 = Q_1 \bar{Q}_0 \bar{Y} + \bar{Q}_1 \bar{Q}_0 Y = \bar{Q}_0 [Q_1 \bar{Y} + \bar{Q}_1 Y] = \bar{Q}_0 [Q_1 \oplus Y]$$

$$K_1 = \bar{Q}_2 \bar{Q}_0 \bar{Y} + Q_2 Q_0 Y = Q_0 [\bar{Q}_2 \bar{Y} + Q_2 Y] = Q_0 (Q_2 \oplus Y)$$

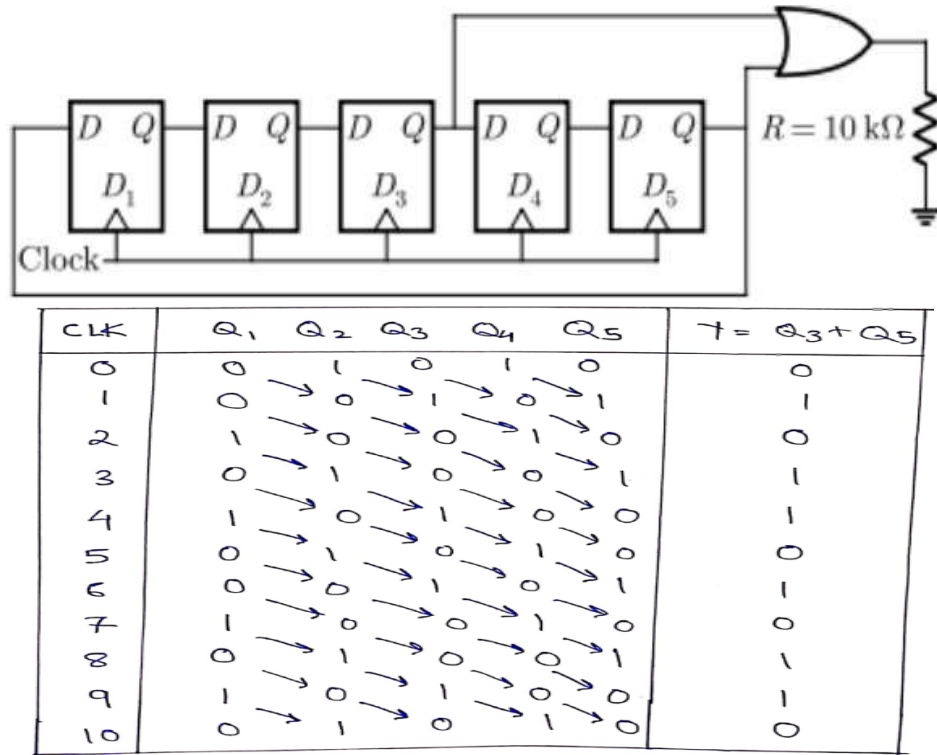
$$K_0 = (\bar{Q}_2 \bar{Q}_1 + Q_2 Q_1) \bar{Y} + (Q_2 \bar{Q}_1 + \bar{Q}_2 Q_1) Y$$

$$= (Q_2 \oplus Q_1) \bar{Y} + (Q_2 \oplus Q_1) Y$$

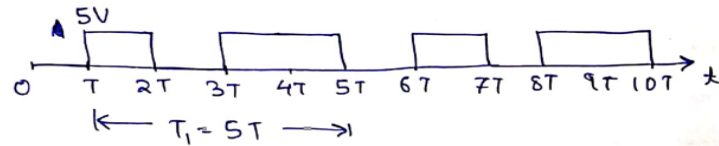
$$= \underline{Q_1 \oplus Q_2 \oplus Y}$$



3. Assume that all the digital gates in the circuit shown in the figure are ideal, the resistor $R = 10\text{ k}\Omega$ and the supply voltage is 5 V . D-flipflops D_1 , D_2 , D_3 , D_4 and D_5 are initialized with logic values 0 , 1 , 0 , 1 and 0 , respectively. The clock has a 30% duty cycle. Find the average power dissipated (in mW) in the resistor R , as well as the output waveform across R .



The waveform at the OR gate o/p is



We can note that the output waveform is periodic with $T_1 = 5T$, where T = one clock period.

$\therefore$ Average Power dissipated in R is :

$$\begin{aligned}
 P &= \frac{V_R^2}{R} = \frac{1}{R} \left[\frac{1}{T_1} \int_0^{T_1} y^2(t) dt \right] \\
 &= \frac{1}{RT_1} \left[\int_T^{2T} 5^2 dt + \int_{3T}^{5T} 5^2 dt \right] \\
 &= \frac{5^2}{RT_1} [(2T - T) + (5T - 3T)] \\
 &= \frac{5^2}{10k \times 5T} \times [T + 2T]
 \end{aligned}$$

$$= \frac{5^2 \times 3T}{10k \times 5T} = \frac{5^2 \times 3}{10k \times 5} = \boxed{1.5 \text{ mW}}$$